

EE5311: Digital IC Design

Tutorial 10

- Input files should be in a directory `designs/<design name>`
- Design directory should be organised as:
 - ❶ `src/*.v`: directory containing verilog files
 - ❷ `src/*.sdc`: clock and other timing specification
 - ❸ `config.json`: configuration file to specify user inputs
 - ❹ `pin_order.cfg`: specify which pins of your top module go to which boundary of the floorplan
- An example adder is in the directory: `/cad/examples/designs`

- Specify the clock period in the Synopsys Design Constraints (SDC) as:

```
set_units -time ns  
create_clock [get_ports <clock port>] -name clock -period <time period>
```

- Example for input port `clk` and period of 5ns:

```
set_units -time ns  
create_clock [get_ports clk] -name clock -period 5.0
```

OpenLane - Pin configuration

- Specify the pins on the boundary of floorplan as:

```
#N
<north pins>
#S
<south pins>
#E
<east pins>
#W
<west pins>
```

- Example configuration for an adder with inputs `a`, `b`, `cin`, and `clk` and output `sum`:

```
#N
a.*
#S
b.*
#E
clk
cin
#W
sum.*
```

Specify various user inputs using the `config.json` file. Example for adder:

```
{
  "//": "Basics",
  "DESIGN_NAME": "adder",
  "VERILOG_FILES": "dir::src/*.v",
  "CLOCK_PERIOD": 5,
  "CLOCK_PORT": "clk",
  "PNR_SDC_FILE": "dir::src/adder.sdc",
  "SIGNOFF_SDC_FILE": "dir::src/adder.sdc",
  "//": "Pin Order",
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg"
}
```

OpenLane - Running the flow

- Run instructions: `openlane.sh <designs directory> <top module>`

- Example:

- ① Copy the designs directory to your path:

```
cp -r /cad/examples/designs .
```

- ② Run: `openlane.sh designs adder`

- ③ Final layout file:

```
designs/adder/runs/auto_pnr/results/signoff/adder.gds
```

- ④ Post-layout STA report:

```
designs/adder/runs/auto_pnr/logs/signoff/30-rcx_mcsta.nom.log
```

- ⑤ Your design meets timing specification if the setup and hold slack are non-negative in the post-layout STA report. e.g.

```
=====
report_worst_slack -max (Setup)
=====
worst slack 0.10
```

```
=====
report_worst_slack -min (Hold)
=====
worst slack 0.33
```

- ⑥ Outputs of various stages of OpenLane flow are in:

```
designs/adder/runs/auto_pnr/results
```